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PUBLICATION

00-README

USAREUR-AF

Maven Smart System (MSS) — Training Curriculum

USAREUR-AF Operational Data Team

HEADQUARTERS
UNITED STATES ARMY EUROPE AND AFRICA
(USAREUR-AF)
Wiesbaden, Germany

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MAVEN SMART SYSTEM (MSS) — TRAINING CURRICULUM

USAREUR-AF OPERATIONAL DATA TEAM

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Distribution: Approved for public release; distribution is unlimited.

PURPOSE

This curriculum provides progressive, doctrine-aligned training for all USAREUR-AF personnel who access, build on, or lead with the Maven Smart System (MSS). It is organized into foundational doctrine publications and platform-specific technical manuals following Army publication conventions.

NEW HERE? START FAST

MSS Quick Start — Get operational in 30 minutes. Read this before anything else if you just got MSS access.

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HOW TO USE THIS CURRICULUM

Step 1 — Know Your Level

Level	You Are...	Start Here
All personnel	A Soldier, officer, or civilian who uses MSS to access operational data	TM-10
All staff, light builders	Building dashboards, forms, or basic pipelines on MSS	TM-20 (after TM-10)
Data-adjacent specialists	Assigned to a data/analytical role (17/25-series, S6/G6, G2/G9)	TM-30 (after TM-20)
Developers / technical specialists	Writing code, building external apps, ML models	TM-40 (track) after TM-30
Senior leaders (O-5+, CSM+)	Commanding or directing a data-capable formation	Data Literacy for Senior Leaders
All — background reading	Anyone who wants to understand data concepts before touching MSS	Data Literacy Technical Reference

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Step 2 — Read the Right Publications

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Senior Leaders ──────────────────────────► Data Literacy for Senior Leaders
                                                  │
                                                  ▼
All Personnel ───────────────────────────► Data Literacy Technical Reference
[recommended]
                                                  │
                                                  ▼
All Personnel ───────────────────────────► TM-10 (Maven User)
                                                  │
                                                  ▼
All Staff ───────────────────────────────► TM-20 (Builder)
                                                  │
                                                  ▼
Data-Adjacent Specialists ───────────────► TM-30 (Advanced Builder)
                                                  │
                                       ┌──────────┴──────────┐
                                       ▼                     ▼
                          TM-40A (ORSA)
                          TM-40B (AI Eng)
                          TM-40C (MLE)
                          TM-40D (PM)
                          TM-40E (KM)
                          TM-40F (SWE)
                                       │
                                       ▼
                          TM-50 series (advanced, per track)
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Step 3 — Reference the Glossary

At any point, consult the Data & Foundry Glossary to translate between general data concepts and MSS/Foundry terminology.

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PUBLICATIONS INDEX

Doctrine Publications

Publication	File	Audience	Purpose
Data Literacy for Senior Leaders	doctrine/DATA_LITERACY_senior_leaders.md	O-5+, CSM+, Senior Civilians	Principles, command responsibilities, evaluating data products
Data Literacy Technical Reference	doctrine/DATA_LITERACY_technical_reference.md	All personnel	Comprehensive data literacy reference — platform-agnostic
Glossary — Data & Foundry Terms	doctrine/GLOSSARY_data_foundry.md	All personnel	Equates general data concepts to MSS/Foundry terminology

Technical Manuals (Platform-Specific)

Publication	File	Audience	Prerequisite
TM-10 — Maven User	tm/TM_10_maven_user/TM_10_MAVEN_USER.md	All staff	None
TM-20 — Builder	tm/TM_20_builder/TM_20_BUILDER.md	All staff	TM-10
TM-30 — Advanced Builder	tm/TM_30_advanced_builder/TM_30_ADVANCED_BUILDER.md	Data-adjacent specialists	TM-10, TM-20

TM-40 Series — Technical Specialist Tracks

Each TM-40 track includes a companion **Concepts Guide** covering background theory and doctrine context for that specialty.

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Publication	File	Concepts Guide	Audience	Prerequisite
TM-40A — ORSA	tm/TM_40A_orsa/TM_40A_ORSA.md	CONCEPTS_GUIDE_TM40A	Operations Research analysts	TM-30
TM-40B — AI Engineer	tm/TM_40B_ai_engineer/TM_40B_AI_ENGINEER.md	CONCEPTS_GUIDE_TM40B	AI/ML specialists	TM-30
TM-40C — Machine Learning Engineer	tm/TM_40C_ml_engineer/TM_40C_ML_ENGINEER.md	CONCEPTS_GUIDE_TM40C	MLEs	TM-30
TM-40D — Program Manager	tm/TM_40D_program_manager/TM_40D_PROGRAM_MANAGER.md	CONCEPTS_GUIDE_TM40D	PMs, resource managers, G8/S8	TM-30
TM-40E — Knowledge Manager	tm/TM_40E_knowledge_manager/TM_40E_KNOWLEDGE_MANAGER.md	CONCEPTS_GUIDE_TM40E	KMOs, 37F, S2/S3/S6 KM roles	TM-30
TM-40F — Software Engineer	tm/TM_40F_software_engineer/TM_40F_SOFTWARE_ENGINEER.md	CONCEPTS_GUIDE_TM40F	SWEs	TM-30

TM-50 Series — Advanced Specialist Tracks

Each TM-50 track includes a companion **Concepts Guide** for advanced theory and doctrine.

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Publication	File	Concepts Guide	Audience	Prerequisite
TM-50A — Advanced ORSA	tm/TM_50A_orsa_advanced/TM_50A_ORSA_ADVANCED.md	CONCEPTS_GUIDE_TM50A	Senior ORSA practitioners	TM-40A
TM-50B — Advanced AI Engineer	tm/TM_50B_ai_engineer_advanced/TM_50B_AI_ENGINEER_ADVANCED.md	CONCEPTS_GUIDE_TM50B	Senior AI engineers	TM-40B
TM-50C — Advanced MLE	tm/TM_50C_ml_engineer_advanced/TM_50C_ML_ENGINEER_ADVANCED.md	CONCEPTS_GUIDE_TM50C	Senior ML engineers	TM-40C
TM-50D — Advanced Program Manager	tm/TM_50D_program_manager_advanced/TM_50D_PROGRAM_MANAGER_ADVANCED.md	CONCEPTS_GUIDE_TM50D	Senior tech PMs	TM-40D
TM-50E — Advanced Knowledge Manager	tm/TM_50E_knowledge_manager_advanced/TM_50E_KNOWLEDGE_MANAGER_ADVANCED.md	CONCEPTS_GUIDE_TM50E	Senior KMOs	TM-40E
TM-50F — Advanced Software Engineer	tm/TM_50F_software_engineer_advanced/TM_50F_SOFTWARE_ENGINEER_ADVANCED.md	CONCEPTS_GUIDE_TM50F	Senior SWEs	TM-40F

TRAINING LEVEL DESCRIPTIONS

TM-10 — Maven User (-10 Level)

Audience: All personnel **Competencies upon completion:** - Access MSS via CAC-based login - Navigate projects, datasets, and applications - Use Workshop applications to consume data and submit forms - Execute authorized Actions - Use Contour and Quiver for basic no-code analysis - Interact with AIP Logic and Agent interfaces - Understand classification markings and authorized export procedures - Troubleshoot common access and application issues

TM-20 — Builder (-20 Level)

Audience: All staff **Competencies upon completion:** - Create and organize Foundry projects via UI - Ingest data using Pipeline Builder (visual, no code) - Create basic Object Types and Link Types via Ontology Manager UI - Create simple Actions via Action Editor UI - Build and publish Workshop

applications with dashboards, forms, and filters - Manage project access and permissions via UI - Use Foundry branching to build and promote via UI - Follow USAREUR-AF naming conventions and builder standards

TM-30 — Advanced Builder (-30 Level)

Audience: Data-adjacent specialists (17/25-series, S6/G6, G2/G9, operational data analysts) **Competencies upon completion:** - Design and build complex multi-page Workshop applications with conditional logic and variable passing - Build advanced Pipeline Builder pipelines with multi-source joins and aggregations (visual) - Design Ontology structures (Object Types, Link Types, Actions) using the UI — architecture thinking, not coding - Conduct advanced analysis in Contour (complex aggregations, pivots, calculated columns, saved views) - Build advanced Quiver dashboards with multi-object analysis and linked views - Configure AIP Logic workflows (not author them) - Review and interpret data lineage graphs - Manage governance workflows with Data Stewards - Manage branching and production promotion via UI - Apply USAREUR-AF C2DAO governance standards and naming conventions

TM-40A — ORSA (-40A Level)

Audience: Operations Research/Systems Analysis officers and NCOs, quantitative analysts **Competencies upon completion:** - Configure Code Workspaces (Python/R) within Foundry - Build statistical models: regression, classification, and validation for readiness/logistics applications - Conduct time series forecasting with ARIMA/SARIMA patterns - Run Monte Carlo simulation for COA comparison and risk quantification - Apply linear programming for resource allocation and scheduling optimization - Design wargame/exercise data collection architecture and aggregation pipelines - Build analytical decision support products (Quiver/Contour) to commander standard - Communicate uncertainty: confidence intervals, sensitivity analysis, briefing standards

TM-40B — AI Engineer (-40B Level)

Audience: AI/ML specialists, data engineers assigned to AI pipeline development **Competencies upon completion:** - Author AIP Logic workflows: prompt engineering, chain design, output handling - Build and configure AIP Agent Studio agents with tools, memory, and orchestration - Implement LLM integration patterns: ontology data grounding, RAG, context construction - Apply AI safety requirements: human-in-the-loop gates, output validation, OPSEC for AI products - Write Python transforms that prepare data for AI consumption - Connect AIP Logic workflows to Object Types and Actions - Test and red-team AI outputs; evaluate quality against defined standards - Deploy and monitor AIP Logic workflows in production

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TM-40C — Machine Learning Engineer (-40C Level)

Audience: ML engineers, data scientists building and deploying models on MSS **Competencies upon completion:** - Configure Code Workspaces for model development (GPU, packages, environment management) - Build and evaluate ML models within the Foundry environment - Manage model versioning, experiment tracking, and reproducibility - Deploy models to production and integrate with Ontology Objects and Actions - Implement MLOps patterns: monitoring, drift detection, retraining triggers - Apply responsible AI practices and model documentation standards for operational use

TM-40D — Program Manager (Technical) (-40D Level)

Audience: Technical project managers leading data, AI, and software capability builds on MSS **Competencies upon completion:** - Apply Agile project management (Scrum/Kanban) to data and AI projects - Write user stories and acceptance criteria for data products - Manage ML/AI project lifecycles from research through production - Translate between technical staff and operational users/commanders - Build sprint tracking and project status dashboards on MSS - Manage technical risk, dependencies, and delivery planning - Define production readiness and Definition of Done for data products - Lead change management and user adoption for new MSS capabilities

TM-40E — Knowledge Manager (-40E Level)

Audience: KMOs, 37F, S2/S3/S6 knowledge management roles, unit S3/S6 knowledge officers **Competencies upon completion:** - Design knowledge architecture for AAR, lessons learned, doctrine, and SOP repositories on MSS - Build AAR capture systems using Workshop forms and Object Type pipelines - Design and operate lessons-learned ingestion and tagging pipelines - Use AIP Logic for knowledge summarization, search augmentation, and theme extraction - Build full-text and semantic search systems over knowledge repositories - Manage doctrine and SOP version control within Foundry - Build personnel expertise mapping (skills/experience registries) - Design knowledge transfer and unit continuity processes using MSS

TM-40F — Software Engineer (-40F Level)

Audience: Software engineers building external applications and platform integrations on MSS **Competencies upon completion:** - Authenticate and query the Foundry Ontology via OSDK (TypeScript/Python) - Execute Actions, subscribe to Object changes, and handle pagination and filtering via OSDK - Use Foundry Platform SDK for dataset operations, file management, and branch management - Build TypeScript Functions on Objects (computed properties, bulk query patterns) - Write and test complex Action validators with TypeScript - Build Slate applications integrated with the Foundry API - Apply USAREUR-AF code review and deployment standards for MSS applications

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TRAINING MANAGEMENT

Administrative Publications

Resource	Location	Audience
Mission Training Plan (MTP)	training_management/MTP_MSS.md	Training managers, S3/G6, commanders
Program of Instruction (POI)	training_management/POI_MSS.md	Training managers, curriculum designers
Course Administrative Data (CAD)	training_management/CAD_MSS.md	Training administrators
Training and Evaluation Outline (TEO)	training_management/TEO_MSS.md	Instructors, evaluators
Annual Training Schedule	training_management/ANNUAL_TRAINING_SCHEDULE.md	Training managers, S3
Enrollment SOP	training_management/ENROLLMENT_SOP.md	Training coordinators
Completion Certificate Template	training_management/COMPLETION_CERTIFICATE.md	Training administrators
Faculty Development Plan	training_management/FACULTY_DEVELOPMENT_PLAN.md	Instructor cadre, S3
Training Policy Letter	training_management/POLICY_LETTER.md	Commanders, training managers
Curriculum Maintenance SOP	training_management/CURRICULUM_MAINTENANCE_SOP.md	Curriculum owner, SMEs, instructors
AAR / Feedback Template	training_management/AAR_TEMPLATE.md	Instructors, evaluators

The MTP provides: TLOs/ELOs per level, individual task lists, Go/No-Go criteria, hour-by-hour blocks of instruction for all levels, training schedule templates, resource requirements, instructor certification standards, practical exercise scenarios (all levels including TM-40), and sustainment requirements.

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Lesson Plans

Resource	Location	Notes
Lesson Plan Template	training_management/lesson_plans/LP_TEMPLATE.md	Blank template for new courses
TM-20 Lesson Plan Outlines	training_management/lesson_plans/TM20_LESSON_PLAN_OUTLINES.md	Builder track
TM-30 Lesson Plan Outlines	training_management/lesson_plans/TM30_LESSON_PLAN_OUTLINES.md	Advanced Builder track
TM-40 Specialist Lesson Plan Outlines	training_management/lesson_plans/TM40_SPECIALIST_LESSON_PLAN_OUTLINES.md	All six TM-40 tracks

Course Syllabi (Student/Instructor-Facing)

Hand the appropriate syllabus to each trainee at the start of their course.

Syllabus	File	Level
TM-10 — Maven User	syllabi/SYLLABUS_TM10.md	All personnel
TM-20 — Builder	syllabi/SYLLABUS_TM20.md	All staff
TM-30 — Advanced Builder	syllabi/SYLLABUS_TM30.md	Data-adjacent specialists
TM-40A — ORSA	syllabi/SYLLABUS_TM40A.md	ORSA analysts
TM-40B — AI Engineer	syllabi/SYLLABUS_TM40B.md	AI/ML specialists
TM-40C — ML Engineer	syllabi/SYLLABUS_TM40C.md	MLEs
TM-40D — Program Manager	syllabi/SYLLABUS_TM40D.md	PMs, G8/S8
TM-40E — Knowledge Manager	syllabi/SYLLABUS_TM40E.md	KMOs, 37F
TM-40F — Software Engineer	syllabi/SYLLABUS_TM40F.md	SWEs

NOTE

TM-50 series syllabi are not yet published. Advanced track instruction is managed through the MTP and TEO. Contact training management for TM-50 course scheduling.

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SUPPORTING REFERENCE MATERIAL

Resource	Location	Notes
MSS Quick Start	QUICK_START.md	30-minute onboarding for new MSS users
Naming & Governance Standards	standards/NAMING_AND_GOVERNANCE_STANDARDS.md	USAREUR-AF MSS naming conventions, governance checklists
MSS Training Hub (web)	mss_info_app/index.html	Standalone HTML training hub — open in browser
Practical Exercises	exercises/	Hands-on exercise packages for TM-10 through TM-40F
Foundry Python Patterns (runnable code)	data_skills/13_foundry_patterns/	Local shim for offline development and testing
Full data science reference library	data_skills/	15 modules: Python → ML → ETL → deployment
SITREP Tracker (applied project)	sitrep_tracker/	Click + Rich + SQLite — reference implementation

QUICK-REFERENCE TRAINING PATHS

For a New Soldier joining the data team:

1. Read Data Literacy for Senior Leaders if O-5+; skip if not
2. Skim Data Literacy Technical Reference Ch 1-3 and Ch 8
3. Complete TM-10 cover to cover
4. Request MSS access; practice navigation tasks
5. If building: complete TM-20
6. If data specialist: complete TM-30 after TM-20
7. If in a specialized role: proceed to the appropriate TM-40 track (see table above)

For an S6 shop standing up MSS capability:

1. Commander reads Data Literacy for Senior Leaders

2. All staff complete TM-10
3. Designate builders: complete TM-20
4. Designate data leads: complete TM-30
5. Reference Glossary throughout
6. Technical specialists proceed to TM-40 tracks per role assignment

By MOS/Role — TM-40 Track Selection:

Role / MOS	Recommended TM-40 Track
17A/17C — Cyber officer/NCO	TM-40F (SWE) or TM-40B (AI Eng)
25D — IT specialist	TM-40F (SWE)
25U — Signal support	TM-40F (SWE) or TM-40D (PM)
FA49 — Operations Research	TM-40A (ORSA)
37F — Psychological Operations	TM-40E (KM)
G2/S2 analyst	TM-40A (ORSA) or TM-40E (KM)
G6/S6 data officer	TM-40F (SWE) or TM-40B (AI Eng)
G8/S8 resource manager	TM-40D (PM)
G9/Civil Affairs	TM-40D (PM) or TM-40E (KM)
Data scientist (GS/contractor)	TM-40A (ORSA) or TM-40C (MLE)
AI/ML engineer (GS/contractor)	TM-40B (AI Eng) or TM-40C (MLE)
KMO / Knowledge Officer	TM-40E (KM)
PM / Program Manager	TM-40D (PM)

USAREUR-AF Operational Data Team — UNCLASSIFIED Version 4.0 | March 2026 | TM-40 and TM-50 series complete | Supersedes previous MAVEN_FIELD_MANUAL.md

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